

Assignment–1

- Q1.** Define Artificial Intelligence (AI). Explain any five real-life applications of AI.
- Q2.** What is Machine Learning (ML)? Differentiate between Traditional Programming and Machine Learning with suitable examples.
- Q3.** Define Feature Engineering. Explain why Feature Engineering is considered the most important step before training a Machine Learning model.
- Q4.** Explain the GIGO (Garbage In, Garbage Out) principle with a suitable real-world example.
- Q5.** Differentiate between: (a) Data and Dataset (b) Feature and Target Variable (c) Training Data and Testing Data.
- Q6.** Why can't Machine Learning models learn directly from raw data? Explain any six common problems found in real-world datasets.
- Q7.** Explain the importance of good features and bad features with suitable examples.
- Q8.** Write short notes on: (i) Supervised Learning (ii) Unsupervised Learning (iii) Reinforcement Learning. Give one real-life application of each.
- Q9.** Explain the statement: "Better features are usually more valuable than more complex algorithms."
- Q10.** Explain the complete workflow of a Machine Learning project from Data Collection to Prediction using a neat diagram.
- Q11.** Create a Pandas DataFrame with columns Student, Hours, Attendance, Marks and Result using the given student records, and display it.
- Q12.** Using the DataFrame, display the first 3 rows, last 2 rows, shape, column names and data types.
- Q13.** Separate the dataset into Features (X) and Target Variable (y) and display both.
- Q14.** Using Pandas, display students with Marks > 60, Attendance >= 80, average Marks and highest Attendance.
- Q15.** Write a short note (150–200 words): "Why should every Data Scientist learn Pandas before Machine Learning?"

Sample Dataset for Q11–Q14

Student	Hours	Attendance	Marks	Result
Rahul	2	70	55	Fail
Priya	6	92	82	Pass
Aman	4	80	68	Pass
Neha	1	60	48	Fail

